

# Preliminary analysis of selected wind turbine SCADA datasets

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## Kelmarsh

The first analysed dataset comes from Kelmarsh wind farm located in the UK. Data was collected using the SCADA system (Supervisory Control And Data Acquisition) from January 1st to June 30th 2021 from six wind turbines. The measurements were averaged over 10-minutes time blocks assuming that the variables were nearly constant during this time.

## Results

The following results were obtained during the first analysis of the data:

|                            | <b>Turbine 01</b>   | <b>Turbine 02</b>   | <b>Turbine 03</b>   | <b>Turbine 04</b>   | <b>Turbine 05</b>   | <b>Turbine 06</b>   |
|----------------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
| <b>Columns (variables)</b> | 299                 | 299                 | 299                 | 299                 | 299                 | 299                 |
| <b>Datapoints</b>          | 26064               | 26064               | 26064               | 26064               | 26064               | 26064               |
| <b>First timestamp</b>     | 2021-01-01 00:00:00 | 2021-01-01 00:00:00 | 2021-01-01 00:00:00 | 2021-01-01 00:00:00 | 2021-01-01 00:00:00 | 2021-01-01 00:00:00 |
| <b>Last timestamp</b>      | 2021-06-30 23:50:00 | 2021-06-30 23:50:00 | 2021-06-30 23:50:00 | 2021-06-30 23:50:00 | 2021-06-30 23:50:00 | 2021-06-30 23:50:00 |
| <b>Missing values</b>      | 564309              | 563890              | 561447              | 567260              | 568249              | 583129              |

This analysis shows a large number of missing values. After taking a closer look at the data, we noticed that some of the columns contain only missing data for all turbines. To be precise, the following columns turned out to be empty:

1. 'Lost Production (Contractual Global) (kWh)',
2. 'Lost Production (Contractual Custom) (kWh)',
3. 'Potential power met mast anemometer (kW)',
4. 'Potential power met mast anemometer MPC (kW)',
5. 'Time-based Contractual Avail. (Global)',
6. 'Time-based Contractual Avail. (Custom)',
7. 'Production-based Contractual Avail. (Global)',
8. 'Production-based Contractual Avail. (Custom)',
9. 'Equivalent Full Load Hours counter (s)'

For this reason, we decided to remove those variables from the dataset, which led to results presented below:

|                                | <b>Turbine<br/>01</b> | <b>Turbine<br/>02</b> | <b>Turbine<br/>03</b> | <b>Turbine<br/>04</b> | <b>Turbine<br/>05</b> | <b>Turbine<br/>06</b> |
|--------------------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| <b>Columns<br/>(variables)</b> | 290                   | 290                   | 290                   | 290                   | 290                   | 290                   |
| <b>Missing values</b>          | 329733                | 329314                | 326871                | 332684                | 333673                | 348553                |

## Penmanshiel

The second analysed dataset comes from a British wind farm located in Penmanshiel. As above, the SCADA system with 10-minutes time blocks was used. Data was collected from January 1st to June 30th 2021 from five wind turbines.

### Results

The preliminary analysis led to following results:

|                            | <b>Turbine 01</b>      | <b>Turbine 02</b>      | <b>Turbine 03</b>      | <b>Turbine 04</b>      | <b>Turbine 05</b>      |
|----------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|
| <b>Columns (variables)</b> | 300                    | 300                    | 300                    | 300                    | 300                    |
| <b>Datapoints</b>          | 26064                  | 26064                  | 26064                  | 26064                  | 26064                  |
| <b>First timestamp</b>     | 2021-01-01<br>00:00:00 | 2021-01-01<br>00:00:00 | 2021-01-01<br>00:00:00 | 2021-01-01<br>00:00:00 | 2021-01-01<br>00:00:00 |
| <b>Last timestamp</b>      | 2021-06-30<br>23:50:00 | 2021-06-30<br>23:50:00 | 2021-06-30<br>23:50:00 | 2021-06-30<br>23:50:00 | 2021-06-30<br>23:50:00 |
| <b>Missing values</b>      | 459162                 | 361192                 | 367388                 | 359515                 | 363038                 |

For this dataset, the nine columns mentioned above also turned out to be empty. After removing them, we obtained the following results:

|                            | <b>Turbine 01</b> | <b>Turbine 02</b> | <b>Turbine 03</b> | <b>Turbine 04</b> | <b>Turbine 05</b> |
|----------------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
| <b>Columns (variables)</b> | 291               | 291               | 291               | 291               | 291               |
| <b>Missing values</b>      | 224586            | 126616            | 132812            | 124939            | 128462            |

## EDP

The next analyzed dataset comes from one of EDP's wind farms. It was collected during the whole 2017 and consists of SCADA signals averaged over 10-minuets time blocks.

### Results

The following results have been obtained:

|                                | <b>Turbine 01</b>            | <b>Turbine 06</b>            | <b>Turbine 07</b>            | <b>Turbine 11</b>            |
|--------------------------------|------------------------------|------------------------------|------------------------------|------------------------------|
| <b>Columns<br/>(variables)</b> | 82                           | 82                           | 82                           | 82                           |
| <b>Datapoints</b>              | 52244                        | 52346                        | 52294                        | 52352                        |
| <b>First<br/>timestamp</b>     | 2017-01-01<br>00:00:00+00:00 | 2017-01-01<br>00:00:00+00:00 | 2017-01-01<br>00:00:00+00:00 | 2017-01-01<br>00:00:00+00:00 |
| <b>Last<br/>timestamp</b>      | 2017-12-31<br>23:50:00+00:00 | 2017-12-31<br>23:50:00+00:00 | 2017-12-31<br>23:50:00+00:00 | 2017-12-31<br>23:50:00+00:00 |
| <b>Missing<br/>values</b>      | 2                            | 2                            | 2                            | 2                            |

It is worth to mention that the low number of missing values obtained here is not reliable. It was obtained by calculating the number of empty fields in the dataset. However, we can notice that the number of datapoints differs for each turbine. In fact, there should be 52.560 datapoints assuming that the turbine works the whole year without breaks. To obtain more reliable result and be able to compare the number of missing values with the previous datasets, we calculated it again including the missing rows with 82 variables each. The results we obtained are presented below:

|                       | <b>Turbine 01</b> | <b>Turbine 06</b> | <b>Turbine 07</b> | <b>Turbine 11</b> |
|-----------------------|-------------------|-------------------|-------------------|-------------------|
| <b>Missing values</b> | 25914             | 17550             | 21814             | 17058             |